

2016.0 RANGE ROVER (LG), 303-01

ENGINE - TDV6 3.0L DIESEL - GEN 2/TDV6 3.0L DIESEL

RIGHT CAMSHAFT (G1509136)

REMOVAL AND INSTALLATION

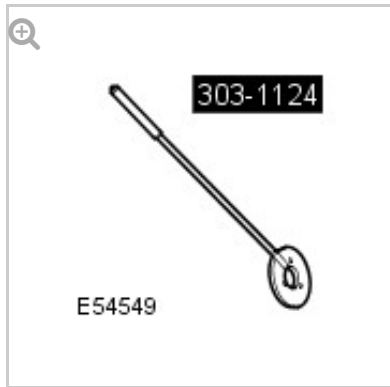
12.13.20	CAMSHAFTS - RH BANK - RENEW	3000 CC, TDV6	8.2	USED WITHINS	+
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SPECIAL TOOL(S)



303-1117

Timing Peg,
Automatic
Transmission



303-1124

Holding Tool,
Camshaft Front
Pulley



303-1126

Timing Peg,
Camshaft Pulley

REMOVAL

1. Disconnect the battery ground cable.
Refer to: [Specifications](#) (414-00 Battery and Charging System - General Information, Specifications).

2. **WARNING:**

Make sure to support the vehicle with axle stands.

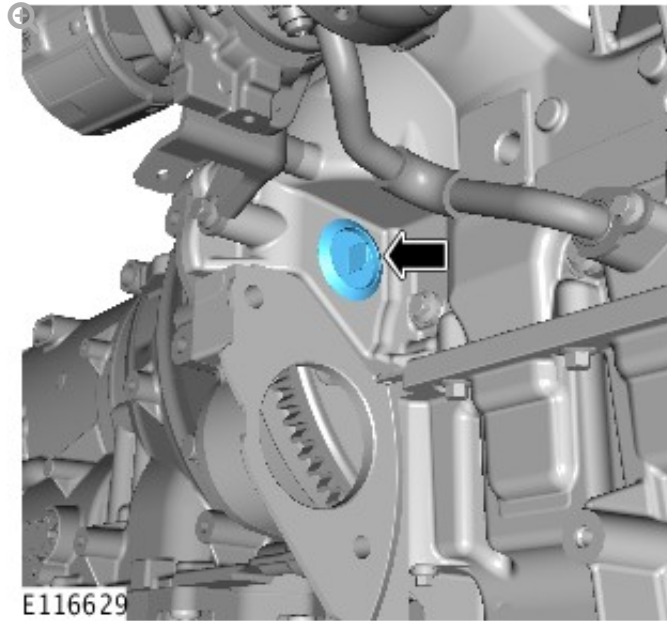
Raise and support the vehicle.

3. Refer to: [Starter Motor](#) (303-06A Starting System - TDV6 3.0L

Diesel, Removal and Installation).

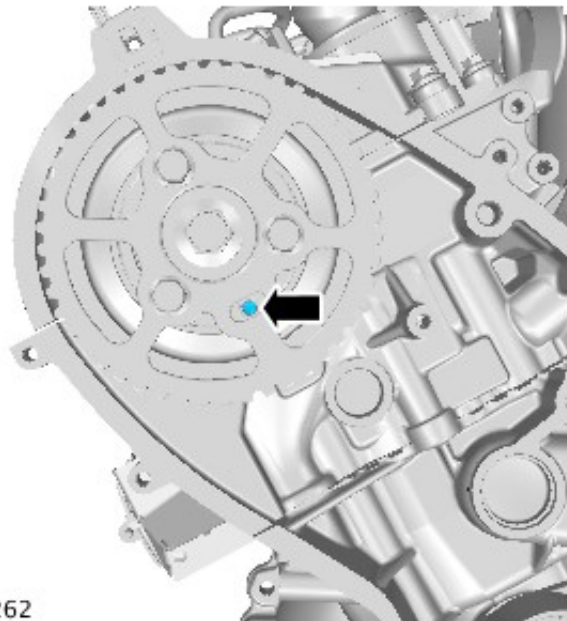
4. Refer to: [Brake Vacuum Pump - TDV6 3.0L Diesel](#) (206-07 Power Brake Actuation, Removal and Installation).
5. Refer to: [Right Valve Cover](#) (303-01F Engine - SDV6 3.0L Diesel - Hybrid Electric Vehicle, Removal and Installation).

6.



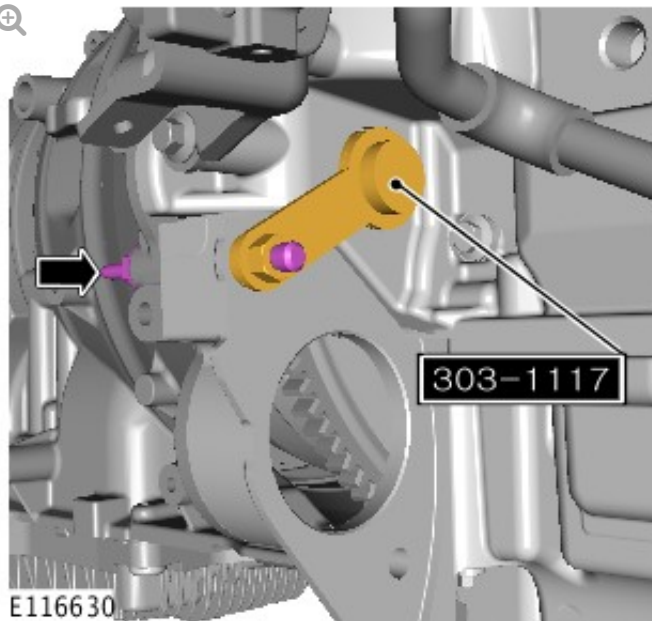
7. Rotate the crankshaft clockwise to align the crankshaft alignment hole in the flywheel or flexplate with the block aperture.

8.



Check the camshaft pulley alignment holes are correctly aligned. If the alignment holes are not aligned, rotate the crankshaft one full turn clockwise.

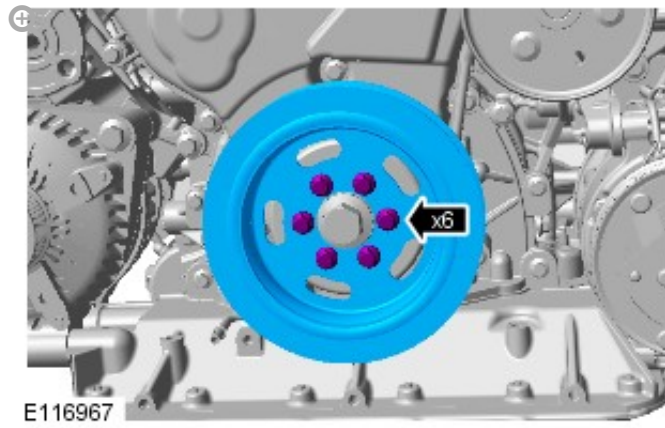
9.



Install the special tool.

Special Tool(s): [303-1117](#)

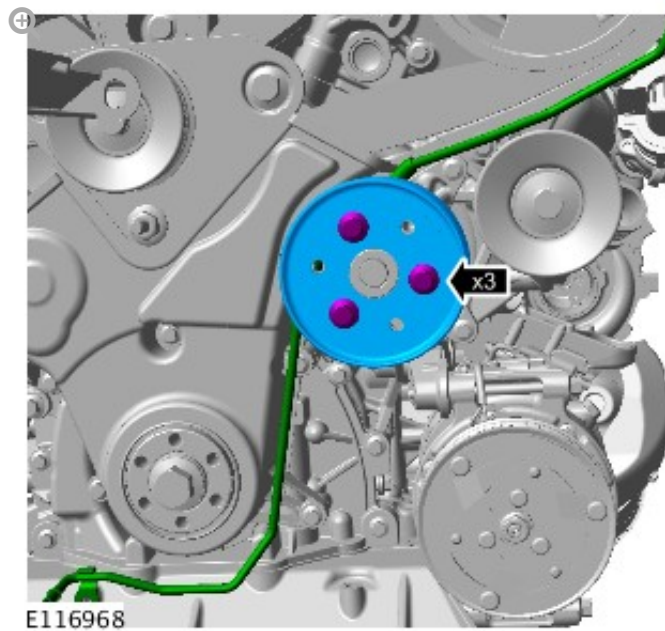
10.



11.

CAUTION:

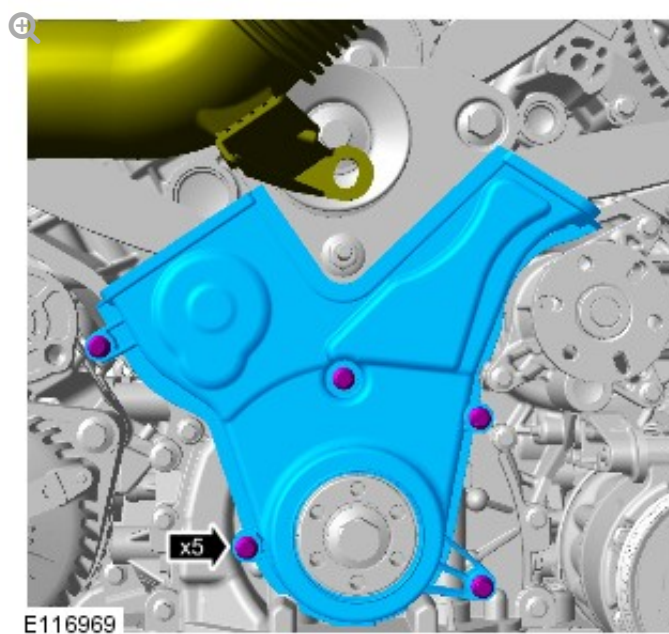
Discard the bolts.



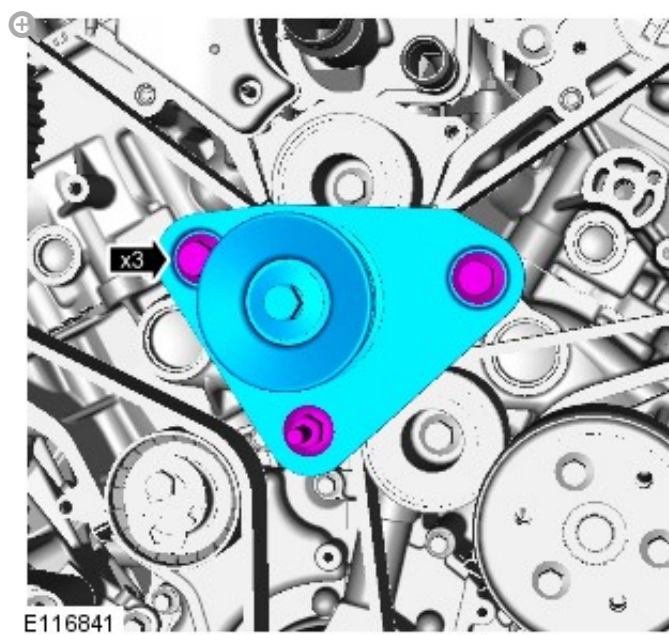
12.

NOTE:

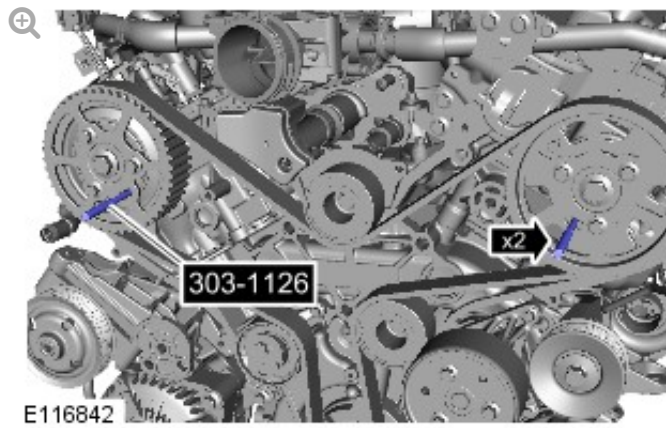
Discard the gasket.



13.



14.



Special Tool(s): [303-1126](#)

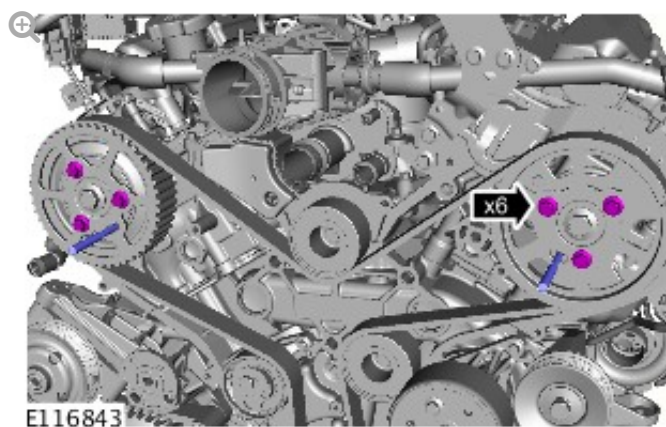
15.

CAUTION:

Do not use the special tools to lock the camshafts. Failure to follow this instruction may result in damage to the engine or the special tools.

NOTE:

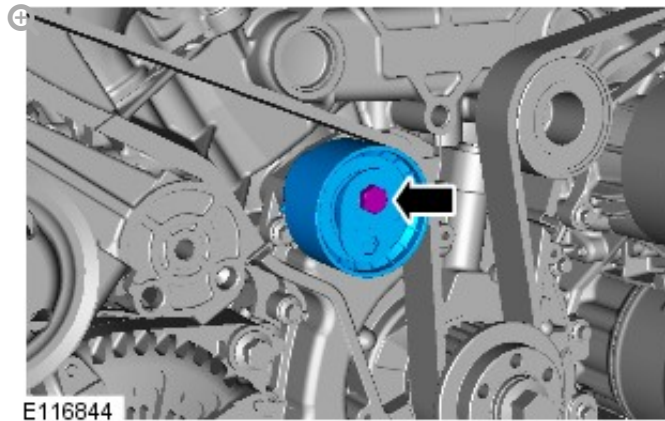
Do not loosen the bolts more than 2 turns.



16.

CAUTIONS:

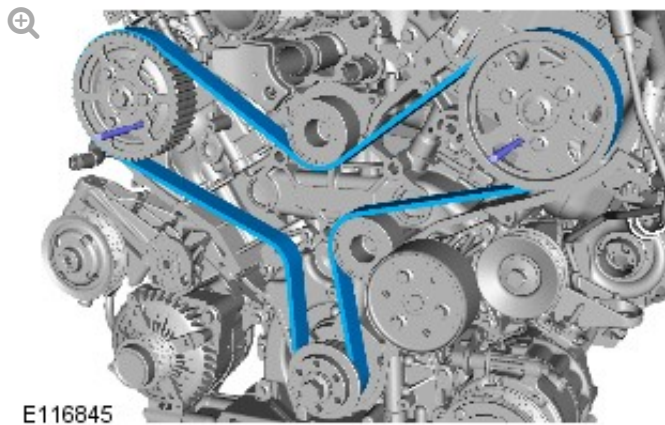
- Discard the component.
- Discard the bolt.



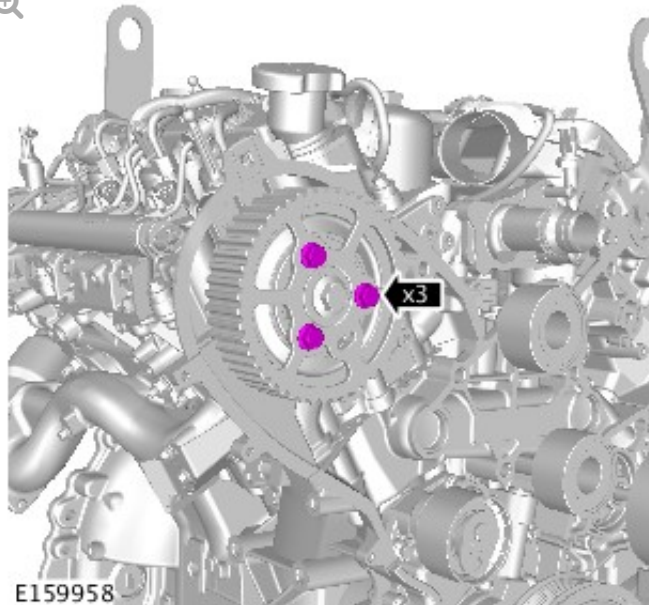
17.

CAUTION:

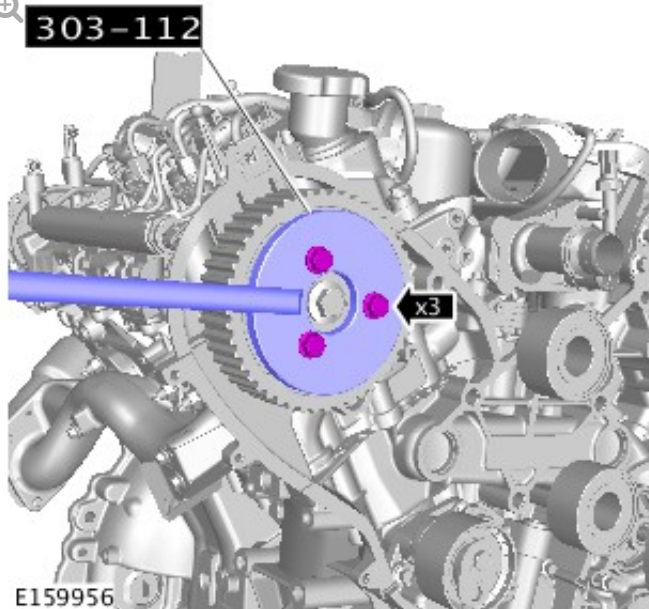
Discard the component.



18.



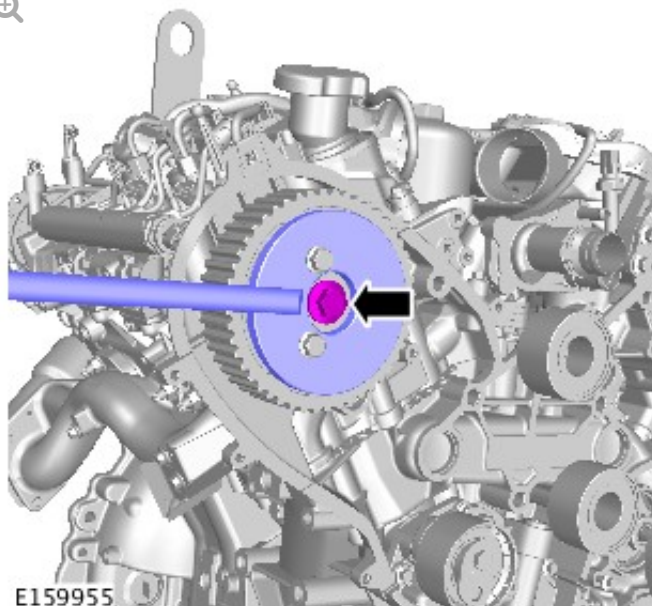
19.



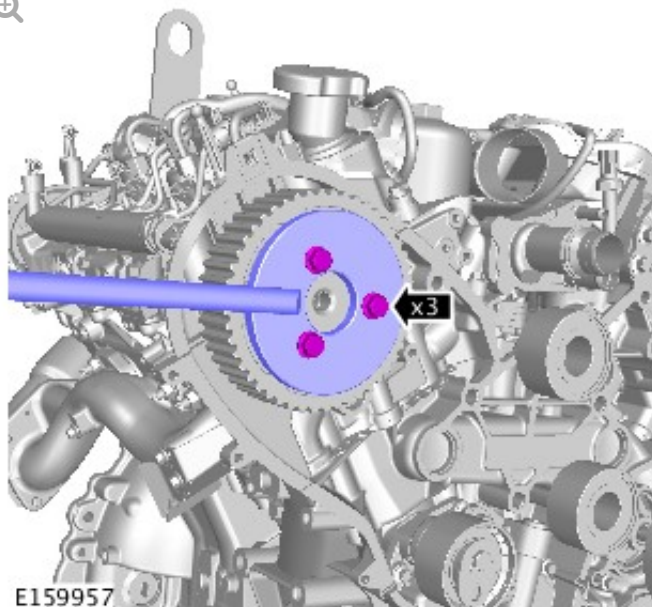
Install the special tool.

Special Tool(s): [303-1124](#)

20.



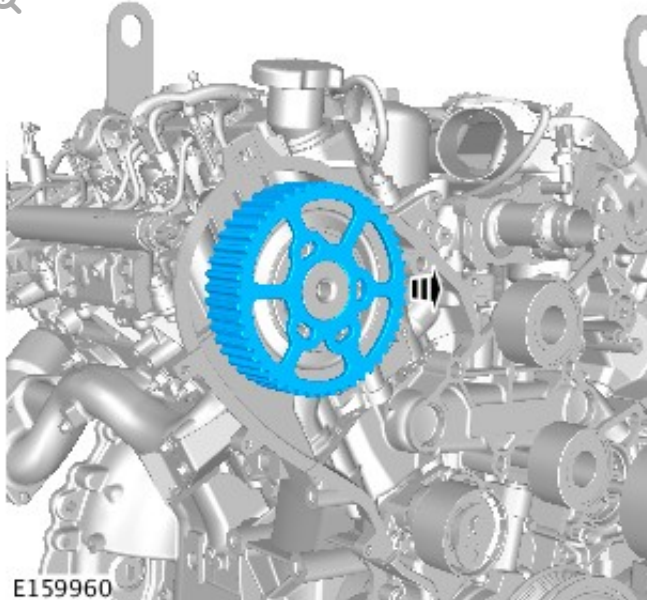
21.



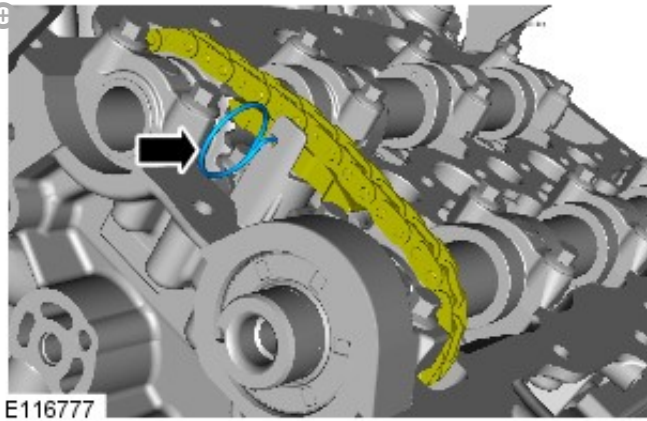
Remove the special tool.

Special Tool(s): [303-1124](#)

22.

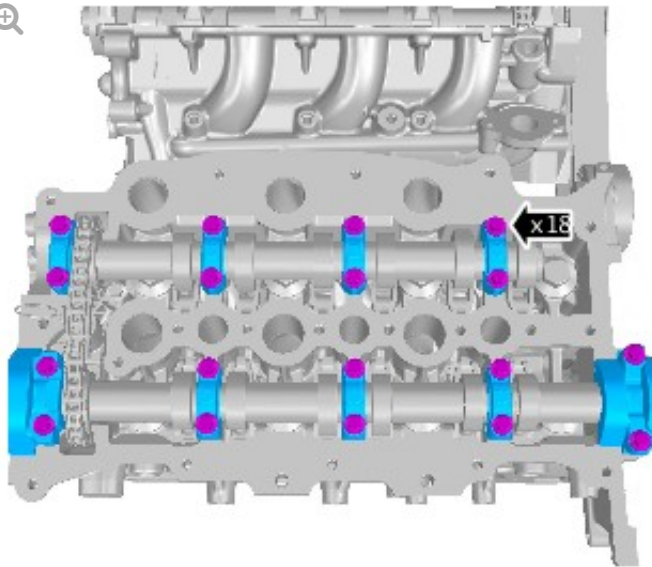


23.



1. Reposition the secondary timing chain tensioner.
2. Retain the secondary timing chain tensioner plunger.

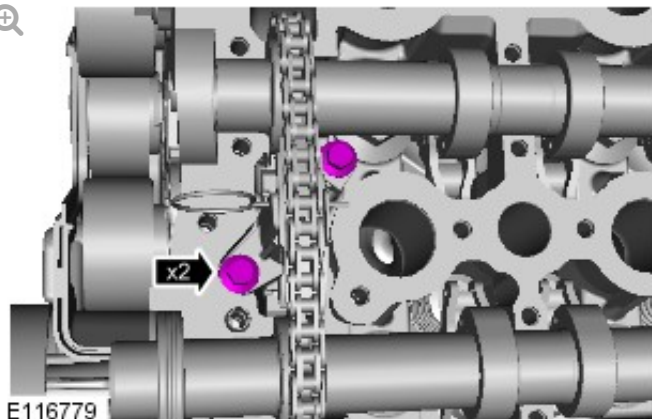
24.



E159959

Remove the camshaft bearing caps evenly.

25.

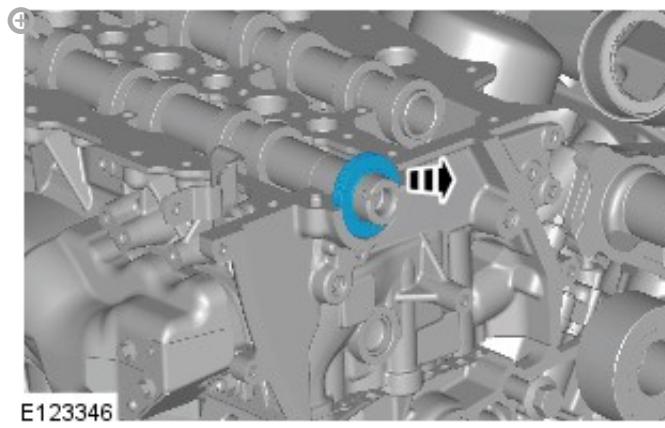


E116779

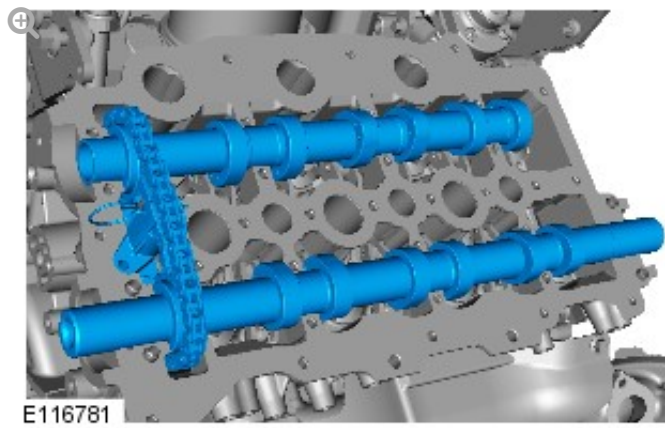
26.

CAUTION:

Discard the seal.

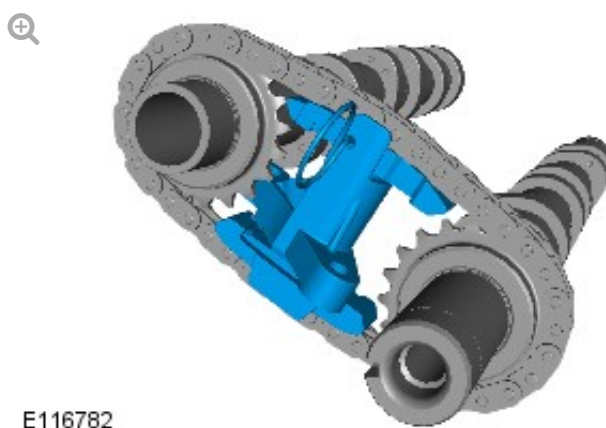


27.



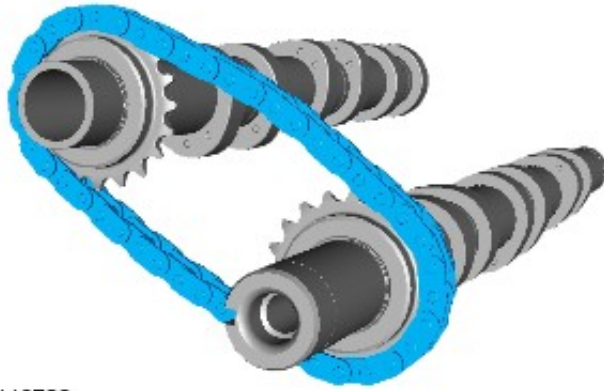
Remove the RH bank camshafts and secondary timing chain tensioner assembly.

28.



Remove the secondary timing chain tensioner assembly.

29.

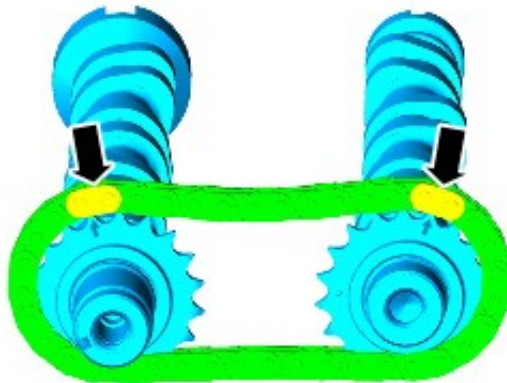


E116783

Remove the secondary timing chain from the camshafts.

INSTALLATION

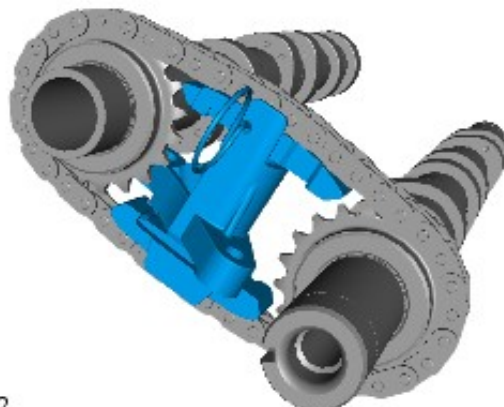
1.



E86461

Install the secondary timing chain onto the camshafts.

2.



E116782

Install the secondary timing chain tensioner assembly.

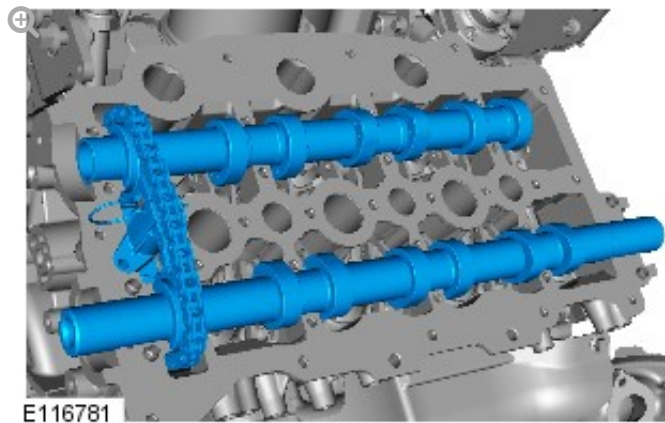
3.

CAUTION:

Make sure that the dots on the camshafts are aligned at the 12 o'clock position. Failure to follow this instruction may result in damage to the engine.

NOTES:

- Lubricate the camshafts and the camshaft bearing caps with oil meeting Jaguar specification prior to installation.
- Use hypoid oil to lubricate the camshafts.

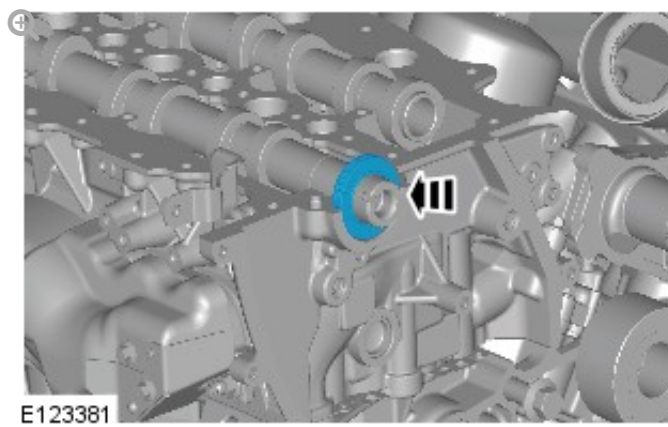


Install the camshafts.

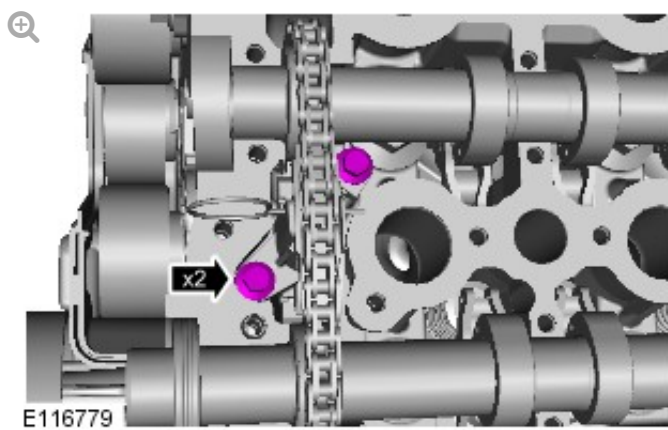
4.

NOTE:

Install a new seal.

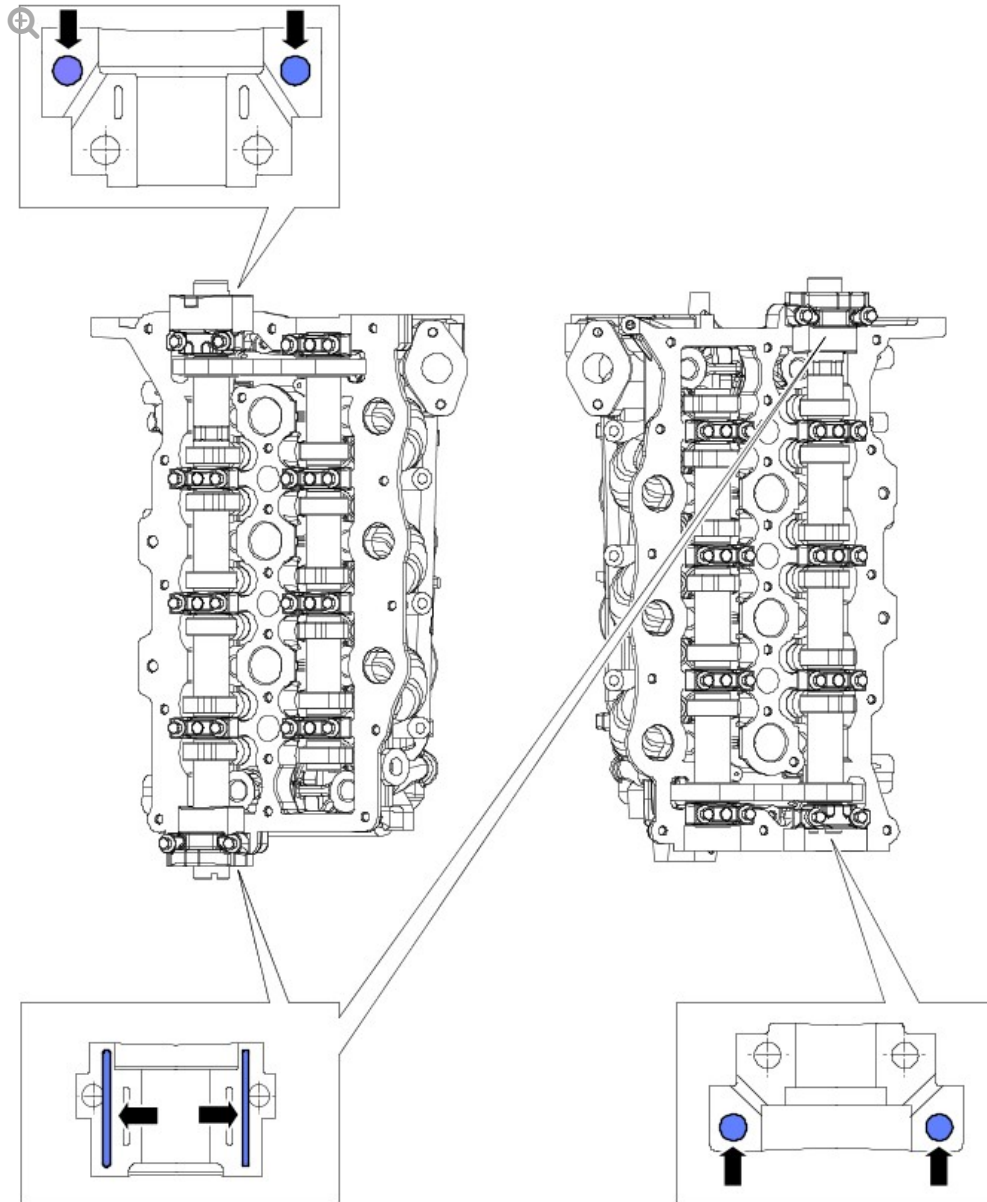


5.



Torque: 10 Nm

6.



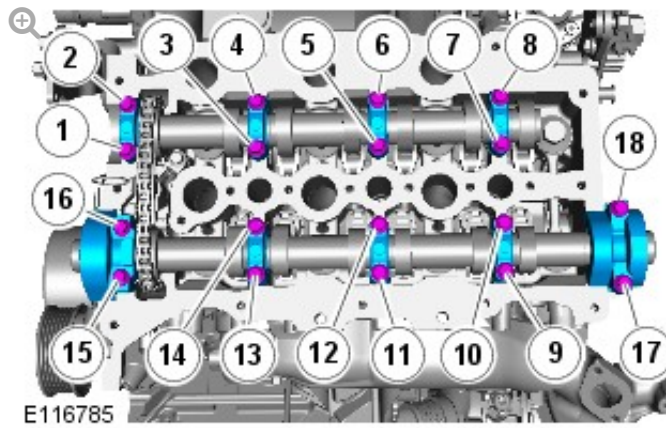
E116784

- Apply Loctite 518 sealant to the exhaust camshaft seal bearing caps.
- Apply Loctite 518 sealant, 2 mm wide, to the LH rear and RH front camshaft bearing caps.
- Apply Loctite 518 sealant, 7 mm diameter, to the LH front and RH rear camshaft bearing caps.

7.

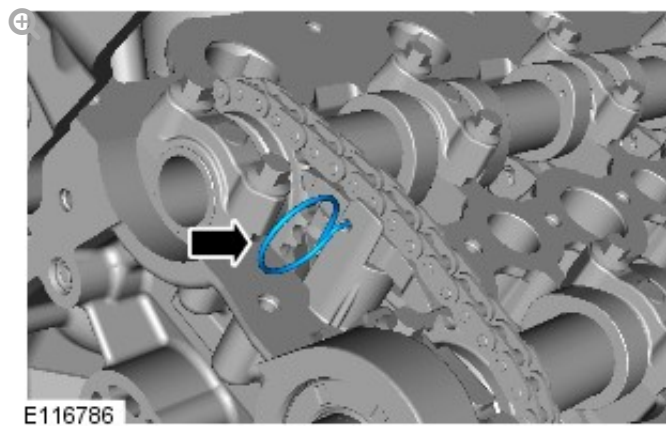
NOTE:

Tighten the bolts in the indicated sequence.



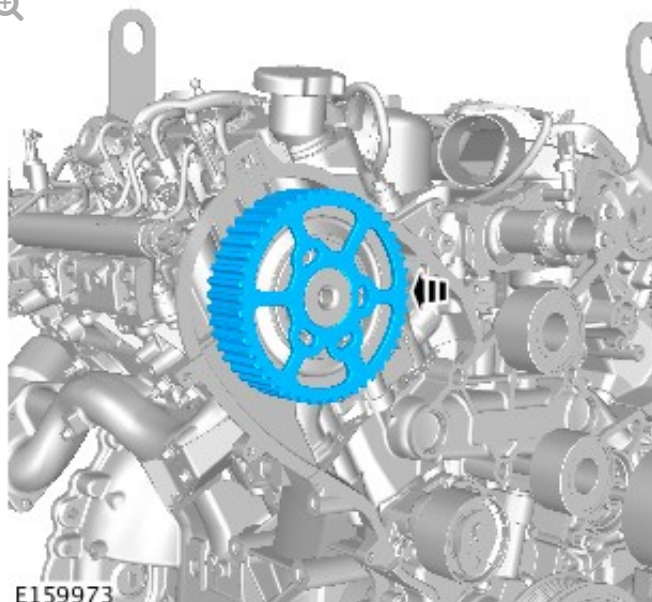
- Install the camshaft bearing caps in their original positions.
- Stage 1: Bolts 1 to 14, 1 Nm.
- Stage 2: Bolts 1 to 14, 5 Nm.
- Stage 3: Bolts 1 to 14, 10 Nm.
- Stage 4: Bolts 15 to 18, 1 Nm.
- Stage 5: Bolts 15 to 18, 5 Nm.
- Stage 6: Bolts 15 to 18, 10 Nm.

8.

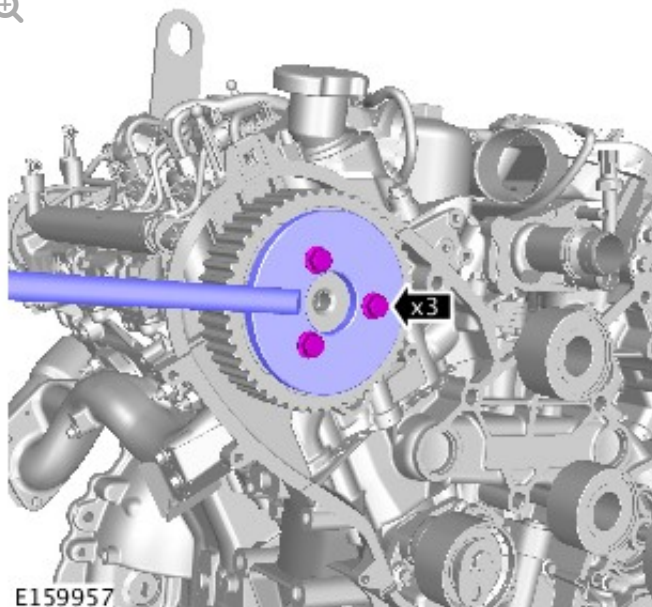


Remove the secondary timing chain tensioner retaining pin.

9.



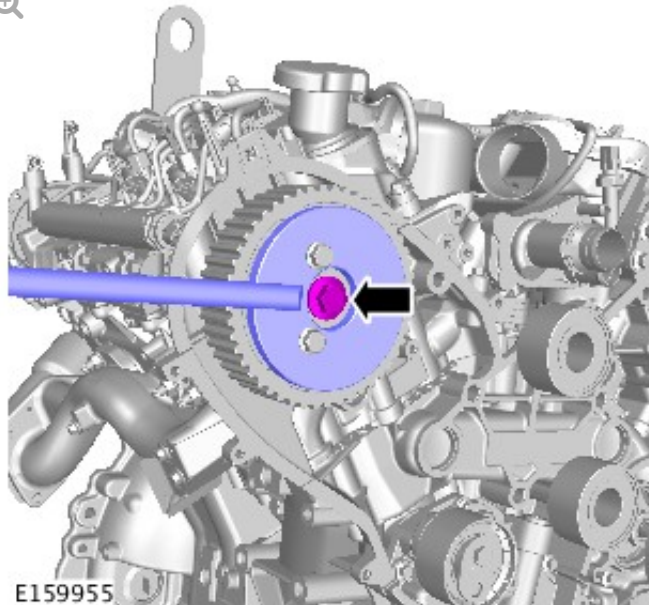
10.



Install the special tool.

Special Tool(s): [303-1124](#)

11.

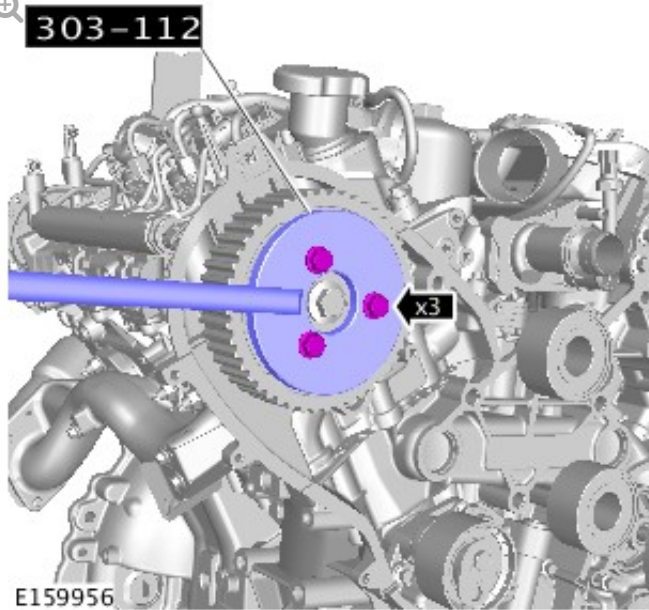


Torque:

Stage 1: **80 Nm**

Stage 2: **80°**

12.



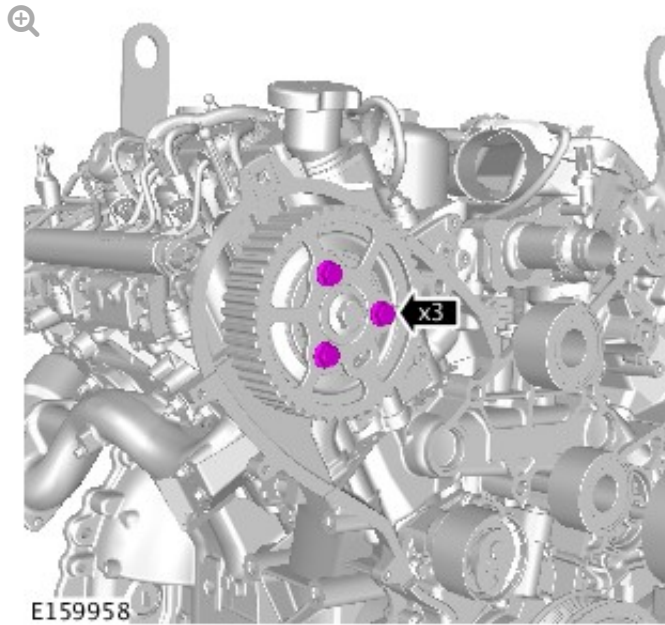
Remove the special tool.

Special Tool(s): [303-1124](#)

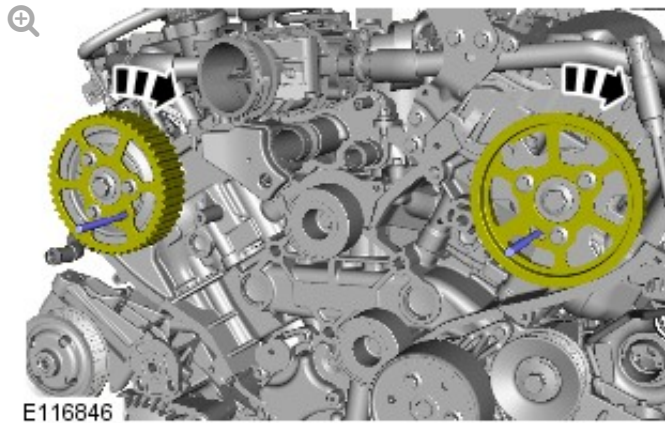
13.

CAUTION:

Only tighten the bolt finger-tight at this stage.



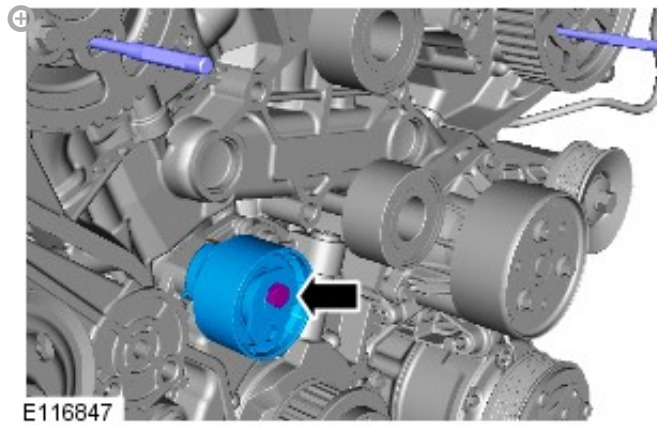
14.



15.

CAUTIONS:

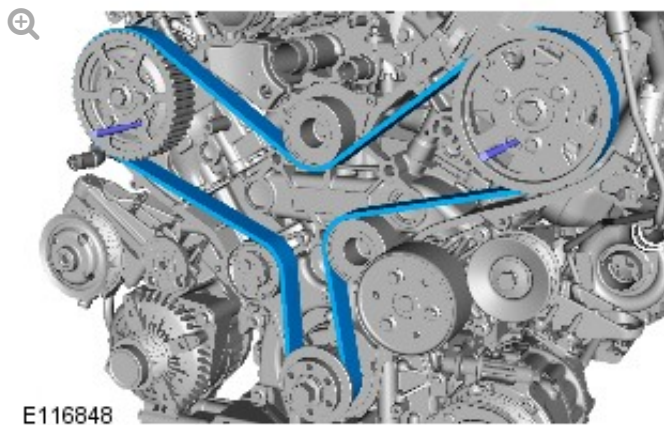
- Make sure that a new bolt is installed.
- Only tighten the bolts finger-tight at this stage.



16.

CAUTION:

Make sure the camshaft pulleys remain in the clockwise position.



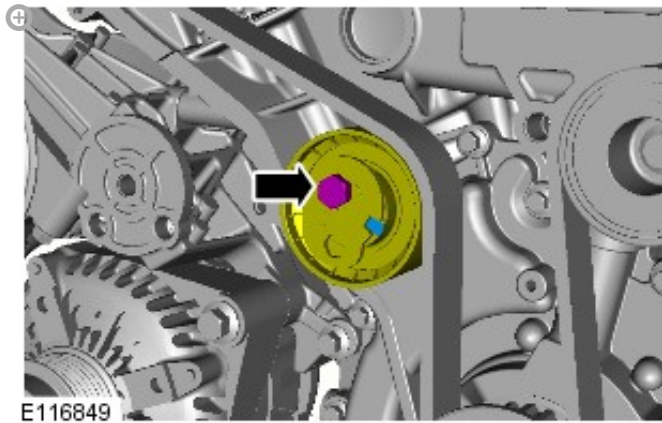
- Install the new timing belt.
- Starting at the crankshaft pulley, install the timing belt in a counter-clockwise direction, in the sequence shown.
- Stage one: Attach the timing belt to the crankshaft pulley.
- Stage two: Attach the timing belt to the idler pulley.
- Stage three: Attach the timing belt to the left-hand camshaft pulley.
- Stage four: Attach the timing belt to the idler pulley.

- Stage five: Attach the timing belt to the RH camshaft pulley.
- Stage six: Attach the timing belt to the timing belt tensioner.

17.

CAUTION:

Make sure the timing belt tensioner window is aligned with the groove as illustrated.

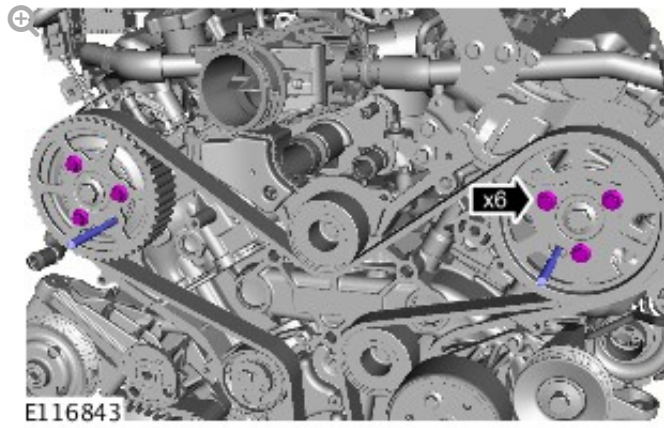


- Tension the timing belt.
- Rotate the tensioner assembly counter-clockwise.
- *Torque: 26 Nm*

18.

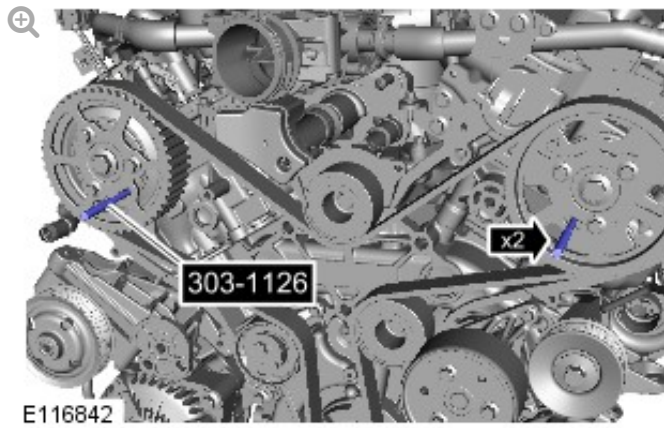
CAUTION:

Do not use the special tools to lock the camshafts. Failure to follow this instruction may result in damage to the engine or the special tools.



- Using a suitable tool, counterhold the camshaft pulley center retaining bolts.
- *Torque: 23 Nm*

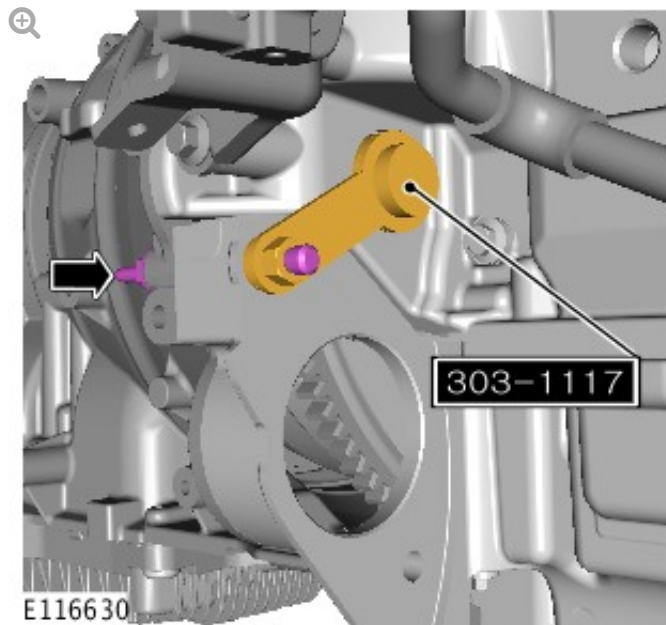
19.



Remove the special tool.

Special Tool(s): [303-1126](#)

20.



Remove the special tool.

Special Tool(s): [303-1117](#)

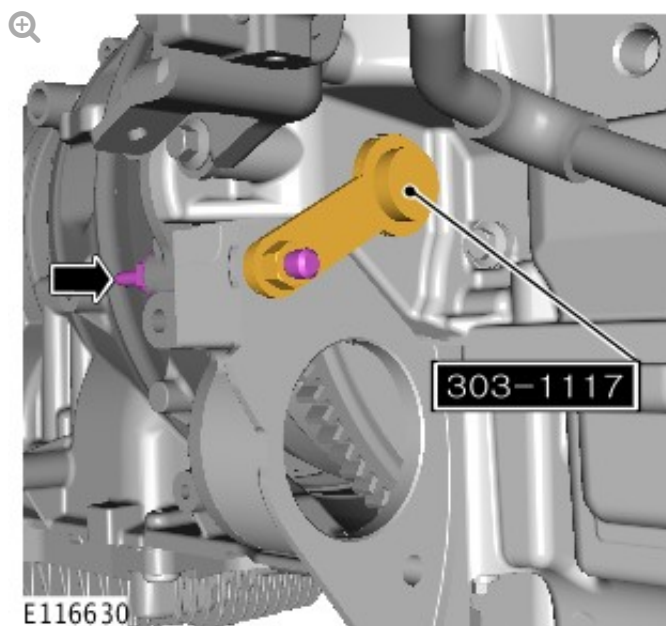
21.

CAUTION:

Only rotate the crankshaft clockwise.

Rotate the engine two complete turns clockwise.

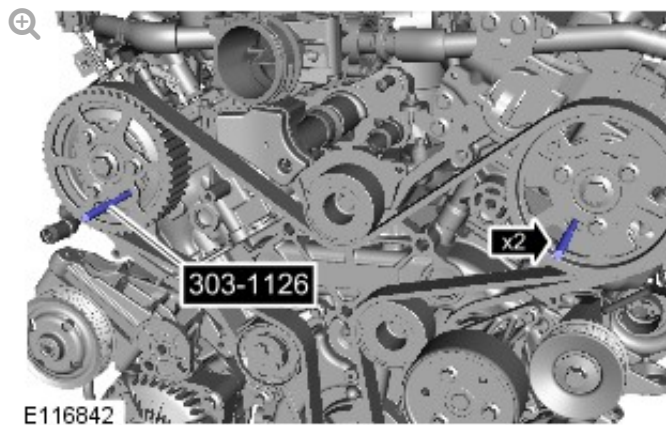
22.



Install with the special tool.

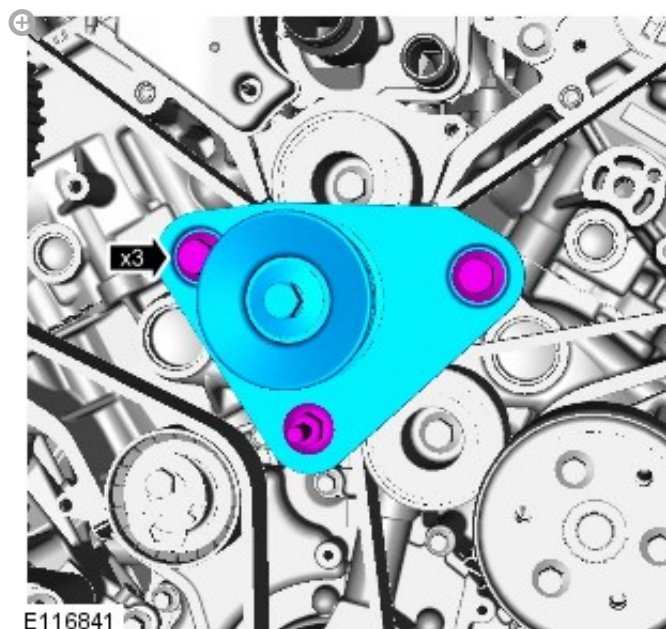
Special Tool(s): [303-1117](#)

23.



- Install the special tools to the exhaust camshaft pulleys.
Special Tool(s): [303-1126](#)
- If the special tool does not fit correctly, repeat the timing belt installation procedure.
- Remove the special tools from the camshaft pulleys.

24.

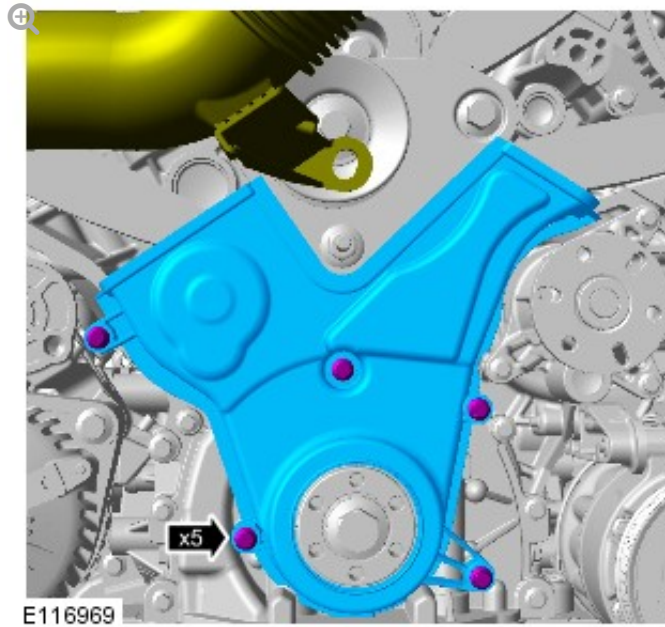


Torque: 80 Nm

25.

NOTE:

Install a new gasket.

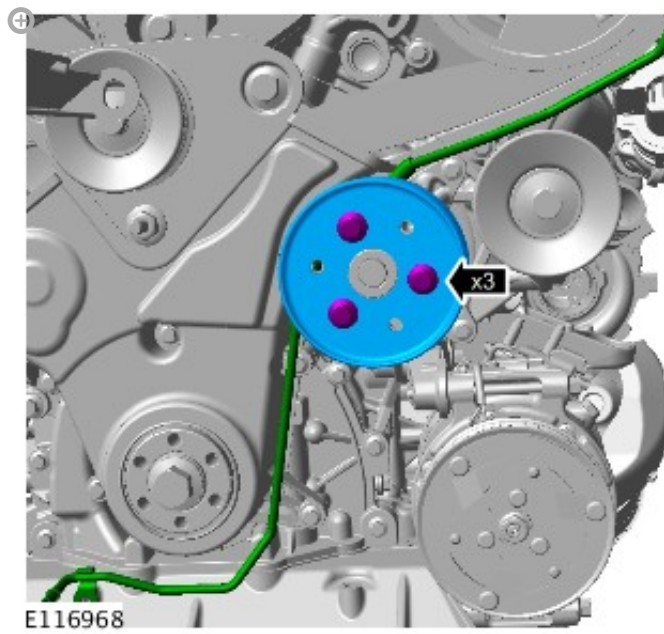


Torque: 10 Nm

26.

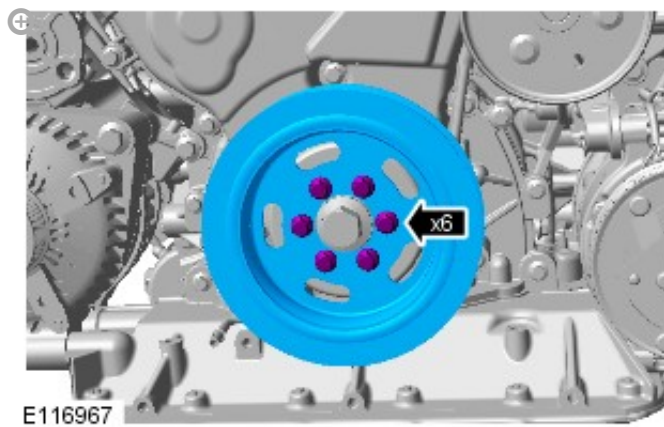
CAUTION:

Make sure that new bolts are installed.



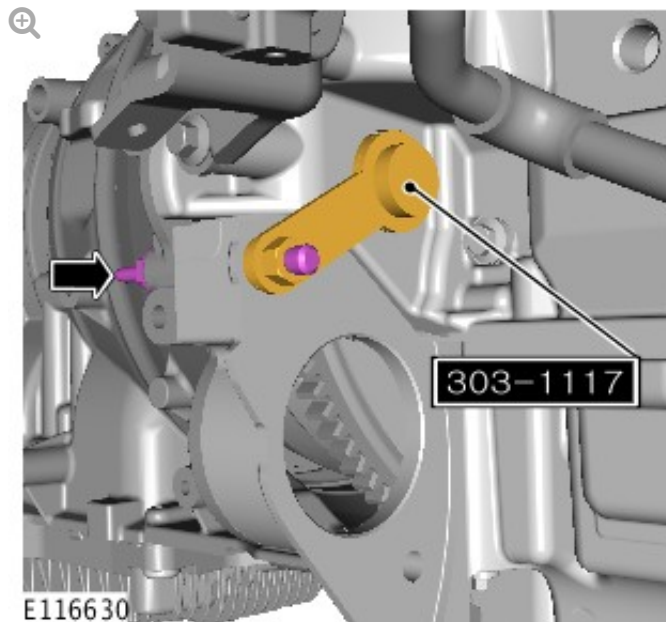
Torque: 24 Nm

27.



Torque: 25 Nm

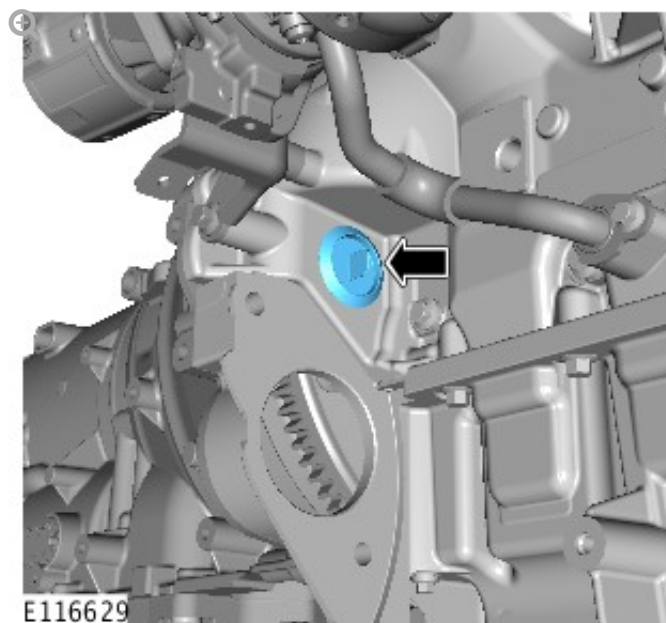
28.



Remove the special tool.

Special Tool(s): [303-1117](#)

29.



30. Refer to: [Right Valve Cover](#) (303-01F Engine - SDV6 3.0L Diesel - Hybrid Electric Vehicle, Removal and Installation).

31. Refer to: [Brake Vacuum Pump - TDV6 3.0L Diesel](#) (206-07 Power Brake Actuation, Removal and Installation).

32. Refer to: [Starter Motor](#) (303-06A Starting System - TDV6 3.0L Diesel, Removal and Installation).
33. Disconnect the battery ground cable.
Refer to: [Specifications](#) (414-00 Battery and Charging System - General Information, Specifications).